# Chapter 9
# Spatiotemporaryism




**Abstract** In this chapter we devise a spacetime logic and argue that temporaryism must give way to spatiotemporaryism, which latter construes variation in what exists as variation across spacetime. In Sect. 9.1 we argue that much of the rationale for thinking, in a prerelativistic setting, that what exists varies across time, should survive the finding that there is no absolute and total temporal order and rationalise the corresponding thought that what exists varies across spacetime. In Sect. 9.2 we introduce a spacetime-sensitive language and a spacetime logic with operators and relations defined over the fourfold causal structure of spacetime. In Sect. 9.3 and Sect 9.4 we use these tools to articulate and compare competing spatiotemporaryist ontologies and contrast them with spatiopermanentism according to which what exists does not vary across spacetime. In Sect. 9.5 we show which relativistic views can naturally be taken to correspond to which classical theories of time.

In this chapter, we explore the implementation of the revisionary strategy. This strategy steers clear of any pretence that relativistic physics may after all cohere with the assumption of a privileged foliation of spacetime into spacelike hypersurfaces. Instead, it seeks to devise alternative versions of classical theories of time that, unlike their prerelativistic predecessors, are consistent with failure of that assumption.

To implement this strategy, we must first come to terms with the fact that, failing such a privileged foliation, there is no way to identify an absolute and total temporal order. In particular, we can no longer assume that there exist times – in any absolute sense in which truths about existence are truths *simpliciter* – and that each such time divides all others exhaustively into those in the past of it and those in the future of it. Instead, we must do justice to the thought that, as Minkowski famously remarked in 1908, 'space by itself, and time by itself, are doomed to fade away into mere shadows, and only a kind of union of the two will preserve an independent reality' (Minkowski 1952: 75). In other words, we must make spacetime-points our basic points of evaluation and appeal to no objective structure of spacetime other than the fourfold division, determined by each such point, between that point itself, those points in the causal past of it, those in the causal future of it, and those in the elsewhere region of it.

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Variation in what exists, if any, must accordingly be understood as variation across spacetime rather than time. Hence, temporaryism must give way to what we might call *spatiotemporaryism*, and in order to fully articulate this generic view, as well as its more specific varieties, we must ultimately use a language that is spacetime- rather than time-sensitive – at least, we are bound to do so if we are intent on identifying metaphysical theories that might, with reason, be conceived as relativistic counterparts to GBT, presentism and dynamic permanentism. This, in turn, makes it mandatory to devise such a language in the first place, equipped with spatiotemporal operators whose behaviour is correspondingly regulated by what we might call a *spacetime logic*. Once such a language and such a logic are to hand, we can go on to formulate and evaluate competing spatiotemporaryist ontologies, and contrast them with the *spatiopermanentist* view according to which, everywhere in spacetime, what exists also exists everywhere else in spacetime.

But before embarking on this enterprise, we must address the more general worry that a variation in the facts across spacetime simply is unintelligible, or at least borders on the absurd – to the extent, namely, that this would have been our verdict in the prerelativistic setting, had it initially been suggested instead that the facts vary across space.

Our plan for this chapter accordingly is as follows. In Sect. 9.1, we address the general worry just rehearsed and attempt to defuse it by further reflecting on the causal structure of the manifold at each spacetime-point. In Sect. 9.2, we give an informal characterisation of a spacetime-point-sensitive language, including a set of spatiotemporal operators, which proves adequate for the description, at each spacetime-point, of the causally structured spacetime manifold. We go on to devise a spatiotemporal logic that mimics, as closely as possible, the tense logic which we introduced in Chaps. 1, 2 and 3. In Sect. 9.3, we then use these resources to distinguish between spatiotemporaryism and spatiopermanentism, and set out those spatiotemporaryist views that are the most natural candidates for succeeding the temporaryist ontologies described in Chaps. 4 and 5. As it turns out, there are two relativistic successor theories for each of presentism and GBT. In Sect. 9.4, we compare the costs and benefits of these four competing ontologies, single out one of each pair of candidates as superior to its respective rival, and conclude with a brief assessment of one of the most striking consequences that the two superior spatiotemporaryist theories share, *viz.* that what exists at a spacetime-point existing at $s$ may not exist at $s$ itself. Appendix 2 offers a semantic characterisation of each of the systems identified. Finally, in Sect. 9.5, we identify two bridge principles that, once conjoined with the relativistic views, yield the corresponding prerelativistic views, given a suitable translation function from the prerelativistic language into the relativistic language. Taken together, these two principles amount to the prerelativistic assumption of an absolute and total temporal order as determining the sole dimension along which facts may be said to vary. The effect of conjoining relativistic theories with these two bridge principles confirms our earlier suspicion that only one of each pair of successor theories, corresponding to GBT and presentism respectively, is a plausible candidate for serving as the relativistic counterpart to the prerelativistic view it succeeds, being most faithful to the latter's rationale. Appendix 3 renders these considerations precise.

## 9.1 Causal Structure and Factual Variation across Spacetime

In Special Relativity (SR) and General Relativity (GR) alike, for each spacetime-point *s*, there is a fourfold, exhaustive division of the manifold into *s* itself and three disjoint regions: the causal past of *s*, the causal future of *s*, and the elsewhere region of *s*. The causal past of *s* is the region of spacetime-points, if any, happenings at which can causally affect what happens at *s*. The causal future of *s* is the region of spacetime-points, if any, happenings at which can be causally affected by what happens at *s*. The elsewhere region of *s*, at last, is the region of spacetime-points, if any, non-causally separated from *s*, happenings at which can neither causally affect, nor causally be affected by, what happens at *s*. The simplified Minkowski diagram in Fig. 9.1 illustrates this causal structure for the special case of SR.[1]

Here, the causal future includes the surface of the light cone represented by the ∨-shaped region, minus *s*; and *mutatis mutandis* for the causal past and the surface of the ∧-shaped region, minus *s*.

This causal structure, though relative to a given spacetime-point, does *not* depend on any further parameter, such as, for instance, a foliation of spacetime or a reference frame. Accordingly, *if* the facts change from spacetime-point to spacetime-point – so that sentences of the form 'Here-now, φ' may be used to express truths *simpliciter* – then, likewise, there will be truths *simpliciter* expressible by sentences of the forms 'Somewhere in the causal past, φ', 'Somewhere in the causal future, φ', and 'Somewhere in the elsewhere region, φ'. Indeed, *if* the facts can change from spacetime-point to spacetime-point, then such changes can only properly be articulated by using sentences of such forms. The pressing question is, however, whether

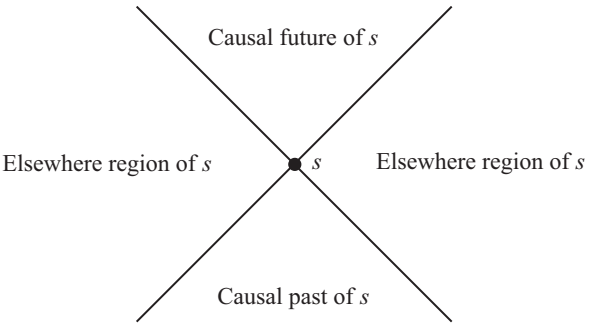



**Fig. 9.1** A spacetime point and its surroundings

[1] The causal structure of Lorentzian manifolds is a topic of its own in relativistic physics. A classic is Chap. 6 of Hawking and Ellis (1973), and a recent survey is provided by Minguzzi and Sánchez (2008).

the thought that the antecedent condition be satisfied is so much as coherent. To fix ideas, and with hindsight, let us focus on facts about what exists.

To begin with note that, for any spacetime-point *s*, what happens at spacetime-points, if any, in the causal past of *s* can still be said to *precede* what happens at *s* – in the very sense of 'precede' at work when we say that causes precede their effects. Similarly, what happens at spacetime-points, if any, in the causal future of *s* can still be said to *succeed* what happens at *s* – in the very sense of 'succeed' at work when we say that effects succeed their causes. That there be variation in what exists across spacetime-points ordered in this way does not become an unintelligible or absurd suggestion, just because what is ordered are spacetime-points rather than time-instants. What *is* puzzling, by contrast, is the finding that when we utter 'now' we refer, if to anything at all, to *here-now*, and that we are accordingly faced with the choice of either denying that there are any spacetime-points besides here-now that causally neither precede nor succeed it, or else accepting that the order in terms of causal precedence and succession is not total and admitting a region of spacetime-points non-causally separated from *here-now*, including here-now, across which there likewise is variation in what exists.

To think, in a relativistic setting, that what exists is relative to spacetime-points – including those, if any, populating the elsewhere region of here-now – is arguably no more counterintuitive than to think, in a prerelativistic setting, that what exists is relative to time-instants. For, arguably, any felt *surplus* of counterintuitiveness merely derives from the assumption that one is in a relativistic setting, and hence from SR and GR themselves. Neither SR nor GR score high on the scale of intuitiveness – which is a good reason not to give too much importance to this scale in evaluating physically informed, metaphysical proposals.

To the extent that one had any temporaryist leanings at all, before one learnt anything about relativistic physics, one thought that what exists may depend on times, but not on places. But this is so because one then thought that there was *absolute simultaneity* between what goes on at distinct places, so that a change of place would not imply a change in what exists, as long as such a spatial change would only take one to a place such that anything occurring there is absolutely simultaneous with what occurs at the departure point. Since in learning about relativistic physics, one learns that there is no such absolute simultaneity, one must now ask oneself afresh whether any objection remains to the suggestion that what exists is relative to spacetime-points. We submit that, for a temporaryist at least, coming to terms with relativistic physics still makes spatiotemporaryism a natural enough idea (Stein 1968; cf. also Stein 1970 and 1991 for further discussion).

For instance, prerelativistically, as a proponent of GBT one might have given a causal interpretation of one's view, thinking that what now exists is exhausted by what now happens and by whatever conspired to make or let it happen, or conspired to make or let happen what so conspired. In other words, one might have thought that what exists is located somewhere in a network of causal chains culminating in what happens now. Variation in what exists at locations upstream and downstream of such causal chains was already contemplated by one's proposal. It then comes as a surprise that 'now' refers, if to anything, to *here-now*, and that what one had

excluded from one's ontology were not just things located beyond the culmination point. Once one realises that there is a region neither upstream nor downstream of those causal chains, and that one's criterion of existence implies that there is also variation in what exists across that region, why should this finding automatically mandate that one abandon that criterion? Is the criterion rendered unintelligible or absurd by this finding?

Alternatively, one might have thought that what now exists is exhausted by what happens now and by whatever has the potential to conspire with what happens now to make or let happen what has not yet occurred. In other words, one might have thought that what exists is located somewhere in a network of causal chains that are going to culminate beyond the here-now in what has not yet happened. Variation in what exists at locations upstream or downstream of such causal chains was already contemplated by one's proposal. It then comes as a surprise that 'now' refers, if to anything, to *here-now*, and that one had included in one's ontology things with the potential to causally affect what is going to happen much later, yet unlike what happens now, without causally affecting anything that is going to happen before – and accordingly had admitted things located in causal chains not including anything that happens now. Once one realises that there is a region neither upstream nor downstream of causal chains including what happens now, and that one's criterion of existence implies that there is also variation in what exists across that region, why should this finding automatically mandate that one abandon that criterion? Is the criterion rendered unintelligible or absurd by this finding?

We submit that the answers to these questions should be negative, because such charges are either based on considerations that give pride of place to everyday intuitions – considerations that should have already been taken to be undermined by science itself, and should anyway not be considered the best currency for evaluating metaphysical proposals – or else because these charges trade on the unwarranted claim that we lack the conceptual resources to properly articulate the spatiotemporaryist positions they seek to discredit. Here, we shall say no more on everyday intuitions. Instead, we will proceed to provide a framework which allows us to properly articulate, in relativistically acceptable ways, the very spatiotemporaryist proposals under attack, and to trace out their respective implications. Providing such a framework should silence any remaining worries that spatiotemporaryism is unintelligible.

## 9.2 Spatiotemporal Logic

There is admittedly no unique phenomenon in natural language, akin to tense, that would allow straightforward recapture by means of spatiotemporal operators, i.e. operators that shift the circumstances of evaluation to those prevailing at certain points in spacetime rather than others. However, we are familiar with ordinary statements to the effect that, say, here and now something is going on that was not going on over there before; and indeed, we are familiar with the idea that any context of

utterance not only determines a time, but also a place – a feature of the context of utterance to which indexicals like 'here' are sensitive. Accordingly, once we buy the Minkowskian lesson that we ultimately cannot treat the conjunctive phrase 'here and now' as a specification of two independent parameter values, but must instead understand it along the lines of the hyphenated 'here-now' denoting a point in relativistic spacetime, we should not find it too hard to conceive of a language whose sentences φ are such that the equivalence '$\varphi \leftrightarrow$ Here-now, φ' is true on each occasion of its use. Just like 'here' and 'now', 'here-now' is an indexical whose value is always determined by the context of utterance. But we can also think up a nonrigid modifier corresponding to this indexical, e.g. 'Locally-presently', and readily conceive of the sentences of the language in question, φ, as being everywhere in spacetime equivalent to 'Locally-presently, φ'. If this were so, we could introduce operators that shift the spacetime-point of evaluation across spacetime, e.g. 'Somewhere in the causal future', 'Everywhere in the causal past', *etc*. – just as, in the prerelativistic setting, we could introduce temporal operators like 'Sometimes in the future' and 'Always in the past'.

In what follows we shall assume the availability of such a spacetime-point-sensitive language, which we model as closely as possible on the tensed language used in previous chapters. However, while in the prerelativistic setting, we had to deal, at each time, with a threefold division of the elements of the temporal order, we must now that we are in a relativistic setting, countenance at each spacetime-point a fourfold division of the spacetime manifold. This means that in order to reach all regions of the manifold, we need more operators than we used to. In particular, we must invoke operators of the form 'Somewhere in the elsewhere region' and 'Everywhere in the elsewhere region', to which no temporal operators correspond.

Let us write

'$\blacktriangle\varphi$' for 'Everywhere in the causal past, φ'
'$\blacktriangledown\varphi$' for 'Everywhere in the causal future, φ'
'$\blacktriangleleft\varphi$' for 'Everywhere in the elsewhere region, φ'
'$\triangle\varphi$' for 'Somewhere in the causal past, φ'
'$\triangledown\varphi$' for 'Somewhere in the causal future, φ'
'$\triangleleft\varphi$' for 'Somewhere in the elsewhere region, φ'
'$\bullet\varphi$' for 'Everywhere in spacetime, φ'
'$\circ\varphi$' for 'Somewhere in spacetime, φ'

We take '$\blacktriangle$', '$\blacktriangledown$' and '$\blacktriangleleft$' as primitive and define the other spacetime operators in their terms as follows. First we define '$\triangle$', '$\triangledown$' and '$\triangleleft$':

$\triangle\varphi \equiv_{df} \neg\blacktriangle\neg\varphi$
$\triangledown\varphi \equiv_{df} \neg\blacktriangledown\neg\varphi$
$\triangleleft\varphi \equiv_{df} \neg\blacktriangleleft\neg\varphi$

Thus, to hold somewhere in the causal past just is not to fail to hold everywhere in the causal past; to hold somewhere in the causal future just is not to fail to hold

everywhere in the causal future; and to hold somewhere in the elsewhere region just is not to fail to hold everywhere in the elsewhere region. Next we define '●' and '○':

$$\bullet\varphi \equiv_{df} \blacktriangle\varphi \,\&\, \varphi \,\&\, \blacktriangleleft\varphi \,\&\, \blacktriangledown\varphi$$
$$\circ\varphi \equiv_{df} \triangle\varphi \vee \varphi \vee \triangleleft\varphi \vee \triangledown\varphi$$

In other words, being everywhere the case just is being the case everywhere in the causal past, here-now, everywhere in the elsewhere region and everywhere in the causal future, while being somewhere the case just is being the case either somewhere in the causal past, or here-now, or somewhere in the elsewhere region, or somewhere in the causal future.

Below, we will give informal glosses on the formal principles we introduce, using labels that wherever possible recognisably correspond to those we used for tense-logical principles. This means that sometimes the numbering does not match the chronological order in which these principles are being introduced. To avoid unnecessary clutter, we will throughout omit the qualification 'causal'.

The following axioms and rules are supposed to govern the aforementioned spacetime operators:

$(A1_R)$ $\quad \varphi \rightarrow \blacktriangle\triangledown\varphi$
$(A2_R)$ $\quad \varphi \rightarrow \blacktriangledown\triangle\varphi$

Whatever is here-now the case is both everywhere in the past somewhere in the future the case, and everywhere in the future somewhere in the past the case. These two axioms mimic the tense-logical axioms (A1), i.e. '$\varphi \rightarrow HF\varphi$', and (A2), i.e. '$\varphi \rightarrow GP\varphi$'.

$(A3_R)$ $\quad \blacktriangle(\varphi \rightarrow \psi) \rightarrow (\blacktriangle\varphi \rightarrow \blacktriangle\psi)$
$(A4_R)$ $\quad \blacktriangledown(\varphi \rightarrow \psi) \rightarrow (\blacktriangledown\varphi \rightarrow \blacktriangledown\psi)$
(AR1) $\quad \blacktriangleleft(\varphi \rightarrow \psi) \rightarrow (\blacktriangleleft\varphi \rightarrow \blacktriangleleft\psi)$

If everywhere in the past $\varphi$ implies $\psi$, then if $\varphi$ is everywhere in the past the case, so is $\psi$; if everywhere in the future $\varphi$ implies $\psi$, then if $\varphi$ is everywhere in the future the case, so is $\psi$; and if everywhere in the elsewhere region $\varphi$ implies $\psi$, then if $\varphi$ is everywhere in the elsewhere region the case, so is $\psi$. $(A3_R)$ and $(A4_R)$ mimic the tense-logical axioms (A3), i.e. '$H(\varphi \rightarrow \psi) \rightarrow (H\varphi \rightarrow H\psi)$', and (A4), i.e. '$G(\varphi \rightarrow \psi) \rightarrow (G\varphi \rightarrow G\psi)$'. The additional axiom (AR1) follows the same pattern.

$(A5_R)$ $\quad \triangledown\triangle\varphi \rightarrow (\triangle\varphi \vee \varphi \vee \triangleleft\varphi \vee \triangledown\varphi)$
$(A6_R)$ $\quad \triangle\triangledown\varphi \rightarrow (\triangle\varphi \vee \varphi \vee \triangleleft\varphi \vee \triangledown\varphi)$
(AR2) $\quad \triangleleft\triangleleft\varphi \rightarrow (\triangle\varphi \vee \varphi \vee \triangleleft\varphi \vee \triangledown\varphi)$

Whatever holds somewhere in the future somewhere in the past, also holds somewhere; whatever holds somewhere in the past somewhere in the future, also holds

somewhere; and whatever holds elsewhere elsewhere, also holds somewhere. The first two of these axioms mimic the tense-logical axioms (A5), i.e. 'FPφ → (Pφ ∨ φ ∨ Fφ)', and (A6), i.e. 'PFφ → (Pφ ∨ φ ∨ Fφ)'. (AR2) has no tense-logical precursor but a similar rationale.

(AR3) $(\lhd\nabla\varphi \vee \nabla\lhd\varphi) \rightarrow (\lhd\varphi \vee \nabla\varphi)$
(AR4) $(\lhd\Delta\varphi \vee \Delta\lhd\varphi) \rightarrow (\lhd\varphi \vee \Delta\varphi)$

If φ holds either elsewhere somewhere in the future, or somewhere in the future elsewhere, then φ either holds elsewhere or somewhere in the future. Similarly, if φ holds either elsewhere somewhere in the past, or somewhere in the past elsewhere, then φ either holds elsewhere or somewhere in the past.

($A7_R$) $\Delta\Delta\varphi \rightarrow \Delta\varphi$

If somewhere in the past, somewhere in the past, φ holds, then somewhere in the past, φ holds. This axiom mimics the tense-logical axiom (A7), i.e. 'PPφ → Pφ'; it prescribes that, in the Kripke-models to be introduced in Appendix 2, the causal precedence relation between spacetime-points is transitive.

(AR5) $\varphi \rightarrow \blacktriangleleft\lhd\varphi$

Whatever is here-now the case is everywhere in the elsewhere region elsewhere the case. (AR5) prescribes that the relation of *being in the elsewhere region of* in those Kripke models is symmetric.

($A14_R$) $\bigcirc\Delta\top$, for $\top$ any chosen tautology

Just as the tense-logical axiom (A14), i.e. 'P⊤ ∨ F⊤', ensures that all models have at least two times standing in relations of precedence, this axiom ensures that all models have at least two spacetime-points standing in relations of causal precedence. As such, it averts the collapse of interestingly different metaphysical theories.

($R1_R$) $\varphi / \blacktriangle\varphi$
($R2_R$) $\varphi / \blacktriangledown\varphi$
(RR) $\varphi / \blacktriangleleft\varphi$

For any theorem φ, it is also a theorem that everywhere in the past, φ is the case, that everywhere in the future, φ is the case, and that everywhere in the elsewhere region, φ is the case. The first two rules mimic the tense-logical rules (R1), i.e. 'φ / Hφ', and (R2), i.e. 'φ / Gφ'. The spacetime-logical rule (RR) follows the same pattern. These three rules allow us to derive the rule which mimics (RD1), i.e. 'φ / Always, φ':

($RD1_R$) $\varphi / \bullet\varphi$

In other words, if φ is a theorem it is likewise a theorem that everywhere, φ is the case. Similarly, the following theorems can now easily be proved[2]:

- $\bullet(\varphi \rightarrow \psi) \rightarrow (\bullet\varphi \rightarrow \bullet\psi)$
- $\bullet\varphi \rightarrow \varphi$
- $\bullet\varphi \rightarrow \bullet\bullet\varphi$
- $\varphi \rightarrow \bullet\circ\varphi$

Accordingly, ● *(everywhere)* and ○ *(somewhere)* behave like □ and ◊ in S5.

In addition to these spatiotemporal operators, we employ spatiotemporal operators of the form '@*m*', abbreviating 'At spacetime-point *m*'. We take 'S' to abbreviate the predicate 'is a spacetime-point' and lay down the following axioms for such spatiotemporal operators which mimic those we postulated for temporal operators of the form 'At *m*':

$(A8_R)$ $@x\varphi \rightarrow Sx$
$(A10_R)$ $@x(\varphi \rightarrow \psi) \rightarrow (@x\varphi \rightarrow @x\psi)$
$(A11_R)$ $@x\neg\varphi \leftrightarrow (Sx \,\&\, \neg@x\varphi)$
$(A12_R)$ $\bullet\varphi \rightarrow (Sx \rightarrow @x\varphi)$
$(A13_R)$ $@x\varphi \rightarrow \bullet@x\varphi$

From $(A12_R)$ and $(RD1_R)$ we have.

$(RD2_R)$ $\varphi \,/\, Sx \rightarrow @x\varphi$

and given $(RD2_R)$, from $(A8_R)$ and $(A10_R)$, we also have.

$(RD3_R)$ $\varphi \rightarrow \psi \,/\, @x\varphi \rightarrow @x\psi$

It follows from $(A11_R)$ and $(A12_R)$, by $(RD3_R)$, that what holds at a spacetime-point, somewhere holds:

$(T2_R)$ $@x\varphi \rightarrow \circ\varphi$[3]

Our theory of quantification and logic of identity remain the same as before, with the sole exception that (A16), i.e. 'Sometimes, E!*m*', is replaced by.

$(A16_R)$ $\circ E!m$

[2] (AR1) is used for proving the first item on this list, axioms (AR3), (AR4) and (AR2) are used for proving the third item, and (AR5) for proving the fourth item. The second item on the list already follows from the definition of '●'.

[3] Suppose $@x\varphi$. Then by $(RD3_R)$, $@x\neg\neg\varphi$. By $(A11_R)$, it follows that $Sx \,\&\, \neg@x\neg\varphi$. Using $(A12_R)$, we get $\circ\varphi$.

In other words, somewhere $m$ exists. Of course, unlike before, we now understand claims of existence and non-existence as being spacetime-point-sensitive.

Given this, we can now immediately add the following axiom that mimics the tense-logical (A22), i.e. '$\forall x(\mathrm{T}x \rightarrow \mathrm{At}\ x, \varphi) \rightarrow \varphi$':

$(\mathrm{A22_R})$ $\quad \forall x(\mathrm{S}x \rightarrow @x\varphi) \rightarrow \varphi \quad$ (with $x$ not free in $\varphi$)

In other words, if it is the case that, for every spacetime-point, at that point $\varphi$ holds, then here-now $\varphi$ also holds. Note that $(\mathrm{A22_R})$ entails that there is at least one spacetime-point: just let $\varphi$ be some logical falsehood, then '$\exists x\mathrm{S}x$' follows.

We take S to be subject to the following constraints:

$(\mathrm{A9_R})$ $\quad \mathrm{S}x \rightarrow \bullet\mathrm{S}x$
$(\mathrm{A23_R})$ $\quad \mathrm{S}x \rightarrow @x\mathrm{E!}x$

That is to say, *being a spacetime-point* is assumed to be a 'ubiquitous' feature of spacetime-points, just as, in the prerelativistic setting, being a time was assumed to be an eternal feature of times: if something is a spacetime-point, then everywhere it is a spacetime-point. Furthermore, any spacetime-point is assumed to exist at itself.

We need the resources to express and attribute spatiotemporal relations between spacetime-points. We take the relations of spatiotemporal location and causal precedence as primitive, symbolize them by '$\textsc{loc}$' and '$\textsc{prec}$' respectively, and begin by laying down the following axioms for '$\textsc{loc}$':

$(\mathrm{A25_R})$ $\quad x \ \textsc{loc}\ y \rightarrow \mathrm{S}y$
$(\mathrm{A26_R})$ $\quad \mathrm{S}x \rightarrow (x \ \textsc{loc}\ y \leftrightarrow x = y)$
$(\mathrm{A27_R})$ $\quad x \ \textsc{loc}\ y \rightarrow \bullet(x \ \textsc{loc}\ y)$

In other words, everything is only ever spatiotemporally located at a spacetime-point; any spacetime-point is located only at itself; and *being spatiotemporally located at* is a 'ubiquitous' feature of things and their locations that does not vary across spacetime: if $x$ is located at $y$, then everywhere, $x$ is located at $y$. We allow spacetime-points to be located at themselves. Naturally, and with hindsight, we do not take $\textsc{loc}$ to be existence-entailing. With $\textsc{loc}$ in place, we can now define what it is to be in spacetime:

(D20) $\quad m$ is in spacetime $\equiv_{\mathrm{df}} \bigcirc\exists x(m \ \textsc{loc}\ x)$

Similarly, we can define what it is to be *a resident of spacetime* as follows:

(D21) $\quad \mathrm{R_{sp}}m \equiv_{\mathrm{df}} m$ is in spacetime $\&\ \bullet\forall x\bullet(m \ \textsc{loc}\ x \rightarrow (\mathrm{E!}x \rightarrow \mathrm{E!}m)) \ \&$
$\bullet(\mathrm{E!}m \rightarrow \exists y(m \ \textsc{loc}\ y))$

The relation of causal precedence among spacetime-points is governed by the following axioms:

$(A28_R)$ $x \text{ PREC } y \rightarrow (Sx \,\&\, Sy)$
$(A29_R)$ $x \text{ PREC } y \rightarrow \bullet(x \text{ PREC } y)$
$(A30_R)$ $\neg(x \text{ PREC } x)$
$(A31_R)$ $(x \text{ PREC } y \,\&\, y \text{ PREC } z) \rightarrow x \text{ PREC } z$

Accordingly, causal precedence is an irreflexive but transitive relation between spacetime-points that does not vary across spacetime. Assuming the irreflexivity of causal precedence amounts to excluding closed causal curves, and hence to the requirement that spacetime satisfies the 'causality condition' (see Hawking and Ellis 1973: 190). By contrast, the transitivity of causal precedence is a fact whatever the spacetime. Like spatiotemporal location, causal precedence is not existence-entailing. PREC is exclusively a relation between spacetime-points. However, with both PREC and LOC in place, we can now define a more general notion of causal precedence among things in time which we symbolize by '$\text{PREC}_g$':

(D22) $m \text{ PREC}_g n \equiv_{df} n$ is in spacetime $\&\, \bigcirc\exists x(m \text{ LOC } x \,\&\, \bullet\forall y(n \text{ LOC } y \rightarrow x \text{ PREC } y))$

We next define the binary relation of *being non-causally separated from*, obtaining between spacetime-points, which we symbolize by 'SEP':

(D23) $m \text{ SEP } n \equiv_{df} Sm \,\&\, Sn \,\&\, \neg(m = n \vee m \text{ PREC } n \vee n \text{ PREC } m)$

Thus, two spacetime-points are non-causally separated just in case they are distinct and none of them causally precedes the other. The relation SEP is irreflexive, but symmetric, and so is not transitive. Like LOC and PREC, SEP is not existence-entailing.

With these notions in place, we can now, lastly, introduce the following four mixed axioms. First consider:

$(A33_R)$ $@x\blacktriangle\varphi \leftrightarrow Sx \,\&\, \bullet\forall y(y \text{ PREC } x \rightarrow @y\varphi)$
$(A34_R)$ $@x\blacktriangledown\varphi \leftrightarrow Sx \,\&\, \bullet\forall y(x \text{ PREC } y \rightarrow @y\varphi)$
(AR6) $@x\blacktriangleleft\varphi \leftrightarrow Sx \,\&\, \bullet\forall y(y \text{ SEP } x \rightarrow @y\varphi)$

where in each case $y$ is not free in $\varphi$ and $x$ and $y$ are distinct variables. That is to say, if at a given spacetime-point $s$, everywhere in the past of that point, $\varphi$ holds, then everywhere it is the case that $\varphi$ holds at any spacetime-point causally preceding $s$, and *vice versa*. If at a given spacetime-point $s$, everywhere in the future of that point, $\varphi$ holds, then everywhere it is the case that $\varphi$ holds at any spacetime-point causally succeeding $s$, and *vice versa*. If at a given spacetime-point $s$, everywhere in the elsewhere region of that point, $\varphi$ holds, then everywhere it is the case that $\varphi$ holds at any spacetime-point non-causally separated from $s$, and *vice versa*. The rationale for these principles is similar in kind to the one we already gave for the tense-logical

axioms (A33), i.e. 'At $x$, H$\varphi \leftrightarrow$ T$x$ & Always, $\forall y(y \prec x \rightarrow$ At $y$, $\varphi)$', and (A34), i.e. 'At $x$, G$\varphi \leftrightarrow$ T$x$ & Always, $\forall y(x \prec y \rightarrow$ At $y$, $\varphi)$'. Finally consider:

$(\text{A24}_\text{R})$ $\quad @x\forall y\varphi \leftrightarrow \bullet\forall y@x(\text{E!}y \rightarrow \varphi)$

where $x$ and $y$ are distinct variables. In other words, if at a given spacetime-point $s$, everything is such that $\varphi$ holds, then everywhere, everything is such that, at $s$, it either does not exist or $\varphi$ holds, and *vice versa*. $(\text{A24}_\text{R})$ corresponds to the tense-logical axiom (A24), i.e. '(At $x$, $\forall y\varphi) \leftrightarrow$ Always, $\forall y$ At $x$, (E!$y \rightarrow \varphi$)'.

Equipped with this spacetime-logic, we now turn to the formulation of different ontological views none of which presupposes the existence of a privileged foliation of spacetime into spacelike hypersurfaces.

## 9.3 Varieties of Spatiotemporaryism

As before in the prerelativistic setting, we can distinguish between two broad classes of views that differ on whether there is factual variation – here: across spacetime rather than time:

$(\text{STA}_\text{R})$ $\quad \forall p, \bullet(\mathscr{T}p \rightarrow \bullet\mathscr{T}p)$
$(\text{DYN}_\text{R})$ $\quad \exists p, \circ(\mathscr{T}p \,\&\, \circ\neg\mathscr{T}p)$

In other words, according to the first type of view, wherever something is true *simpliciter*, and so fully articulates a fact, it also is everywhere else a truth *simpliciter*, while according to the second type of view, somewhere some truth *simpliciter* somewhere else fails to be a truth *simpliciter*.

We continue to make the following two assumptions about truth *simpliciter*:

$$\mathscr{T}\varphi \rightarrow \varphi$$
$$\text{E!}m \rightarrow \mathscr{T}\text{E!}m$$

Given these assumptions, $(\text{STA}_\text{R})$ can be shown to imply *spatiopermanentism*, i.e. the view according to which everywhere in spacetime, whatever exists also exists everywhere else in spacetime:

(SPER) $\quad \bullet\forall x\bullet\text{E!}x$

Thus, assume $(\text{STA}_\text{R})$. Given the two assumptions above, we can derive '$\bullet\forall x\bullet$(E!$x \rightarrow \bullet$E!$x$)'. Thanks to $(\text{A16}_\text{R})$, we can then derive '$\bullet\forall x\circ\bullet\text{E!}x$', and hence, given $\circ\bullet\varphi \rightarrow \bullet\varphi$, '$\bullet\forall x\bullet\text{E!}x$'.

Spatiopermanentism contrasts with *spatiotemporaryism*, i.e. the view according to which somewhere in spacetime, something exists that somewhere else in spacetime does not exist:

(STEMP) $\quad \circ\exists x\circ\neg\text{E!}x$

There is one variety of spatiotemporaryism that pretty straightforwardly corresponds to presentism as we had defined it in Chap. 5, *viz.* the combination of the following two principles:

$(P2_R)$ $Sx \rightarrow @x\blacktriangle\neg E!x$
$(P3_R)$ $Sx \rightarrow @x\blacktriangledown\neg E!x$

In other words, at any spacetime-point $s$, it is the case that everywhere in the causal past of $s$, and everywhere in the causal future of $s$, $s$ fails to exist. Accordingly, at $s$, $s$ is 'one-off' along the trajectory of any particle passing through $s$.

For want of a better label, let us call this view *spatiopresentism*. That spatiopresentism is a version of spatiotemporaryism can be shown as follows:

*Proof*: By $(P2_R)$ and $(A23_R)$, $Sx \rightarrow @x\exists y\blacktriangle\neg E!y$ is a theorem. By $(A22_R)$, then, $\exists y\blacktriangle\neg E!y$ is a theorem. Given $(A14_R)$, i.e. $\bigcirc\triangle\top$, we can infer $\bigcirc\exists y(\blacktriangle\neg E!y \,\&\, \triangle\top)$. Given that $(\blacktriangle\varphi \,\&\, \triangle\top) \rightarrow \bigcirc\varphi$ is a theorem, we can infer $\bigcirc\exists y\bigcirc\neg E!y$. □

Spatiopresentism, as characterised, is as yet neutral as to whether the elsewhere region of here-now is populated. Accordingly, we may add one or the other of the following principles to the mix, obtaining two different versions of spatiopresentism:

(PO) $Sx \rightarrow @x\blacktriangleleft\neg E!x$
(BO) $Sx \rightarrow @x\blacktriangleleft E!x$

The version of spatiopresentism that results from adding (PO) to $(P2_R)$ and $(P3_R)$ entails that everywhere in spacetime there is one and only one spacetime-point:

(1) $E!x \,\&\, E!y \,\&\, Sx \,\&\, Sy \rightarrow x = y$

*Proof:* Assume $(P2_R)$, and $(P3_R)$ and (PO). Next assume $E!m \,\&\, E!n \,\&\, Sm \,\&\, Sn$. From the second assumption, by $(A22_R)$, it follows that there is a spacetime-point, $s$, such that $@sE!m$ and $@sE!n$. From the first and second assumptions and axioms $(A8_R)$, $(A11_R)$, $(A33_R)$, $(A34_R)$ and (AR6), both $@sE!m \rightarrow s = m$ and $@sE!n \rightarrow s = n$ follow. By the transitivity of identity, $m = n$ follows. □

Although, to the best of our knowledge, this version of spatiopresentism has never been defended in print, it has been critically discussed by Hinchliff (2000: 579) and Savitt (2000: 567–68). If here-now is the only spacetime-point that there is, it follows that every resident of spacetime is located here-now.

Another version of spatiopresentism results, once we add (BO) to $(P2_R)$ and $(P3_R)$. This kind of spatiopresentism allows for the elsewhere region of here-now to be populated, i.e. for there to exist a spacetime-point non-causally separated from here-now, and hence equally, for there to exist residents of spacetime that are located in the elsewhere region but are not located here-now.

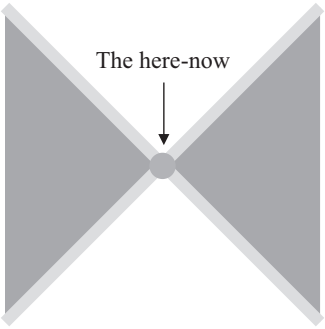



**Fig. 9.2** Bow-tie spatiopresentism

This is the kind of view that was put forward, and to a certain extent defended, by Weingard (1972). Focusing on SR, Weingard writes:

> Consider a space-time point P with its associated lightcones. […] Assuming I-now is real and at P, I should conclude that since I can consider it to be occurring now, any event outside my lightcone can be considered real. But while distant simultaneity is a matter of convention, being real, I take it, cannot be merely a matter of convention. Thus, if an event can be considered real it must be real and so all events outside the lightcone of P (of me-now) are real.
>
> Now in terms of actual physical or experimental facts, it is the class of events that can be considered simultaneous to an event at P, and not the class of events absolutely simultaneous to the events at P, that plays the role in special relativity that the class of events simultaneous to P plays in Newtonian space-time. In each they are the class of events that are not causally connectable with P. And while the class of events simultaneous to an event at P, with respect to some frame of reference, is not a relativistic invariant, the class of events that can be considered simultaneous to events at P is such an invariant. It is just the class of events outside of P's lightcone. Thus, I conclude that the belief that all things that exist now (or are in the present) are real is also true for relativistic space-time, with one qualification. In special relativity the absolute present of an event at P is not the class of events simultaneous with the event with respect to some frame of reference but rather it is the class of events located outside of P's lightcone (plus the events at P). (Weingard 1972: 120-21)

If we understand the regions in dark grey to be those that, according to the view at hand, may be populated, then this version of spatiopresentism can be represented by Fig. 9.2.

By contrast, the spatiopresentist view that results from adding (PO) instead of (BO) would simply have a darkened here-now. We may accordingly call the first version of spatiopresentism *pointy spatiopresentism*, and call the second version *bow-tie spatiopresentism*. We will discuss the relative merits, and relative shortcomings, of these views in Sect. 9.4.

The relativistic counterpart to GBT is naturally conceived to consist in the combination of the following two principles, one of which it shares with spatiopresentism:

$(\mathrm{P1_R})$ $\quad \mathrm{E!}x \rightarrow \blacktriangledown \mathrm{E!}x$

$(\mathrm{P2_R})$ $\quad \mathrm{S}x \rightarrow @x\blacktriangle\neg \mathrm{E!}x$

In other words, everywhere everything everywhere in the causal future still exists, while for any spacetime-point *s*, at *s*, everywhere in the causal past of *s*, *s* did not yet exist. Accordingly, at *s*, *s* is 'new' on any particle's trajectory passing through *s*, while it continues to exist on this trajectory even after the latter has passed through *s*. Let us call this kind of view *relativistic GBT*.

That relativistic GBT is a version of spatiotemporaryism has in effect already been established: the earlier proof that spatiopresentism is a version of spatiotemporaryism only made use of ($P2_R$), which it shares with relativistic GBT, and not ($P3_R$), which relativistic GBT rejects.

On relativistic GBT, as on spatiopresentism, there are, at spacetime-point *s*, no spacetime-points in the causal future of *s*, and hence neither any residents of spacetime that, in the generalised sense of causal precedence, are preceded by *s*. Unlike spatiopresentism, however, relativistic GBT contends that, at *s*, there are spacetime-points in the causal past of *s*, and so things in time that, in the generalised sense of precedence, precede *s*.

Nothing has so far been said about the status of the elsewhere region; and we can, as before, distinguish between two versions of relativistic GBT, a pointy and a bow-tie version. Again, the difference between pointy and bow-tie versions depends on which of the following two principles is accepted:

(PO) $Sx \rightarrow @x\blacktriangleleft\neg E!x$
(BO) $Sx \rightarrow @x\blacktriangleleft E!x$

With (PO) being added to ($P1_R$) and ($P2_R$), we can derive that, at any given spacetime-point, the elsewhere region is unpopulated:

(2) $Sx \rightarrow @x\neg\exists y(x \text{ SEP } y)$

*Proof:* Assume S*x* and assume for *reductio* $\neg@x\neg\exists y(x \text{ SEP } y)$. Then by ($A11_R$), $@x\exists y(x \text{ SEP } y)$. (AR6) allows us to derive the theorem: $x \text{ SEP } y \,\&\, @y\blacktriangleleft\varphi \rightarrow @x\varphi$. Given (PO), we then derive: $x \text{ SEP } y \,\&\, \rightarrow @x\neg E!y$. But then we get $@x\exists y@x\neg E!y)$. Using ($A24_R$) we can get: $\bigcirc\exists y@x(E!y \,\&\, @x\neg E!y)$. Given the logic of @, this yields: $\bigcirc\exists y\bigcirc(E!y \,\&\, \neg E!y)$, which cannot be the case. □

The pointy version of relativistic GBT can be represented by Fig. 9.3.

This kind of view was defended by Stein (1968) in response to charges by Rietdijk (1966) and Putnam (1967). Focusing on SR, Stein maintains that the following characterisation of what exists – or in his parlance, of what is real – 'can be taken over word for word from prerelativistic to Einstein-Minkowski space-time; because the relation "being in the past of" is an ordering of space-time, the obvious compatibility requirement – that if *a* has already become for *b*, and *b* for *c*, then so has *a* for *c* – is satisfied':

> For an event – a man considering, for example – at a space-time point *a*, those events, and only those, have already become (real or determinate), which occur at points in the topological closure of the past of *a*. (Stein 1968: 14)

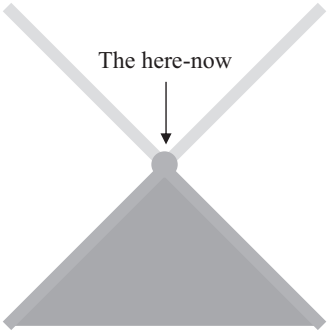



**Fig. 9.3** Pointy relativistic GBT




**Fig. 9.4** Bow-tie relativistic GBT

He goes on to clarify, in a footnote:

> 'Topological closure' because it is natural to say that something in or on the boundary of the past of *a* has 'by now' become for *a*; note that *a* itself is thereby included (and, for Einstein-Minkowski space-time, besides the past of *a* only *a* itself). (Stein 1968: 14)

By contrast, to the best of our knowledge, the bow-tie version of relativistic GBT has never been defended in print. It results from adding (BO) to ($P1_R$) and ($P2_R$) and can be represented by Fig. 9.4.

It is this view that, as we shall argue in the next section, is the best version of relativistic GBT, just as bow-tie spatiopresentism is superior to its pointy rival. Before we ultimately address these issues, however, let us briefly pause to reflect what *dynamic spatiopermanentism* might look like. As a version of spatiopermanentism, it holds that, everywhere, everything exists everywhere else, and so everywhere takes all regions of the manifold to be populated. However, as a version of ($DYN_R$), it nonetheless claims that some propositions are somewhere true *simpliciter* without everywhere being true *simpliciter*.

In Chap. 5, we identified what we considered the best version of dynamic permanentism as the kind of view that, for some specific *p*, claims

Ephemerally, *p*

where 'Ephemerally, φ' was defined as 'Sometimes, (φ & H¬φ & G¬φ)', and where truths about how far from the present, *p* is the case, were supposed to be truths *simpliciter*:

$$D_k p \rightarrow \mathcal{T} D_k p$$

Accordingly, as *p*'s being the case is successively less and less future, and then present, and then more and more past, reality itself changes on this view.

Now, while we could easily redefine 'Ephemerally, φ' as '○(φ & ▲¬φ & ◀¬φ & ▼¬φ)', we would need to use a metric tensor in order to express distances from here-now across spacetime. While we will not here be concerned with fleshing out the details, the overall shape of dynamic spatiopermanentism should be clear enough.

## 9.4 The Intransitivity of Being

We noted before that on the view we called *pointy spatiopresentism*, every resident of spacetime is located here-now. It might be suggested that this commits its proponents to the implausible claim that no resident of spacetime is spatiotemporally extended, i.e. that every resident of spacetime must be point-sized, which would render the view untenable. However, this objection must remain unconvincing. Mereological fusions, if they exist, exist whenever, and wherever, one of their parts exist. Thus, even on the view under attack, residents of spacetime may, for all that, be mereological fusions of spatiotemporal parts, as long as one of their spatiotemporal parts is located here-now. Accordingly, pointy spatiopresentism can after all leave room for the existence of things in spacetime that are spatiotemporally extended.

However, there is another objection, very similar to the one just reviewed, that should carry more conviction (cf. Savitt 2000: 568). In fact, this more powerful objection likewise counts against the pointy version of relativistic GBT, i.e. the kind of view defended by Stein (1968).

Let us, for the sake of argument, engage in the fiction that an embodied consciousness can occupy a single spacetime-point. Here-now, I perceive my different limbs. I furthermore perceive my different limbs to be located at distinct spacetime-points. It is true that, in order for me to perceive my limbs, light signals must reach me, which takes time. So strictly speaking, I only ever perceive that, somewhere in the immediate causal past, my limbs occupy distinct spacetime-points (cf. Russell 1912: Chap. 3). Unlike here-now, my causal past is a region big enough to comprise several spacetime-points and so has the 'thickness' that past location of my different limbs requires. However, according to pointy views, *at no point* in my causal past are there sufficiently many points in order for all my different limbs to be located, not even if we think of each of my limbs as a mereological fusion of residents of

spacetime, where such fusions are located wherever any of their parts are located. Since light signals may simultaneously reach me from different points in my causal past, this need not yet be a problem for pointy views. However, I have every reason to believe that my limbs have persisted. In order for them to have done so, they would have to here-now exist, and so would some points in the causal future of the points in my causal past where I here-now perceive them to have been located. For, *ex hypothesi*, my limbs are fusions of residents of spacetime, and wherever the latter exist, so do points at which they are located. But my different limbs are not all located here-now; so where are they located, if they here-now still exist? It is no consolation to be told that at least on pointy relativistic GBT, as opposed to pointy spatiopresentism, my limbs still exist, for while persistence implies existence, existence is quite compatible with a failure of having persisted – in the sense of persistence at issue here, *viz.* that of continuing to exist while located.

Note that bow-tie views face no comparable problem, because both on bow-tie spatiopresentism and bow-tie relativistic GBT, whenever I here-now perceive my limbs to have been located at points in the causal past of here-now, there exist, here-now, several spacetime-points in the causal future of those points. We take these considerations to disclose a clear advantage of bow-tie views over their pointy rivals.

However, it has been argued that both varieties of bow-tie spatiotemporaryism run afoul of certain principles that are either claimed to be independently plausible or claimed to encode intuitions that these views are meant to do justice to. Thus, Savitt (2000) argues against views such as Weingard's on the following grounds:

> First, it seems reasonable to require that [event] *E* itself be a member of any set *S* of events representing the present for *E* in [Minkowskian spacetime]. Second, it also seems reasonable to require that, if some set of events *S* represents the present for event *E*, then no events in *S* should be in each other's absolute past or absolute future (that is, it should not be the case that all observers at *E* agree that one of the events is, say, earlier than the other). Let us call this the requirement that any set of events representing the present in [Minkowskian spacetime] must be *achronal*. (Savitt 2000: 567)

Savitt then argues that on a view such as bow-tie spatiopresentism, this requirement is violated, because on this view, there may be two events in the elsewhere region of here-now such that one is located at $s$ and the other is located at $s'$, where $s$ PREC $s'$ (see Fig. 9.5).

Savitt's requirement is formulated in terms of the notion of presentness; and his whole discussion is premised on the idea that relativistic counterparts to presentism must rehabilitate that notion within a relativistic setting. Once the revisionary strategy is adopted, this premise becomes somewhat contentious. However, Savitt's requirement is also presented as if it was a general principle not tied to any metaphysical theory. But then it should likewise be a plausible requirement for spatiopermanentists; and it is far from clear how this can be so, unless spatiopermanentists too are under the obligation to rehabilitate the notion of presentness. The only candidate notion of presentness, acceptable to the spatiopermanentist, would seem to be the notion of being located at the spacetime-point that is the referent of 'here-now'. But once presentness is understood along these lines, bow-tie spatiopresentism

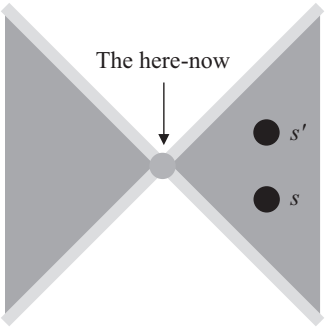



**Fig. 9.5** Illustration of Savitt's objection

would seem to face no problem in its attempt to meet the requirement. Of course, one cannot then, at the same time, oblige proponents of bow-tie spatiopresentism to subscribe to the claim that, everywhere, whatever exists is present *in this sense*; for this would be tantamount to the flat-out assertion that spatiopresentism must be pointy.

Perhaps a better way with Savitt's objection would be to construe it as saying that spatiopresentists *qua* spatiopresentists should be committed to

(3) $\quad \mathrm{E!}x \,\&\, \mathrm{E!}y \,\&\, \mathrm{S}x \,\&\, \mathrm{S}y \rightarrow \neg(x \text{ PREC } y)$

It is clear that (3) is not theory-neutral, since spatiopermanentism and both versions of relativistic GBT reject it. Given its commitment to (1), pointy spatiopresentism obviously vindicates (3). The interesting question is whether spatiopresentists *qua* spatiopresentists are committed to (3). However, failing (PO), the *reductio* proof of (3), based on the two spatiopresentist principles ($\mathrm{P2_R}$) and ($\mathrm{P3_R}$) and mimicking the proofs of (1) and (2) as closely as possible, must at some stage appeal to the principle

(BO+) $\quad \mathrm{E!}x \rightarrow \blacktriangleleft \mathrm{E!}x$[4]

which, given (AR1), (AR5) and (RR), is equivalent to '$\triangleleft \mathrm{E!}x \rightarrow \mathrm{E!}x$'.[5] However, (BO+) is much stronger than (BO), and so it is questionable why bow-tie spatiopresentists should be committed to it. We will comment on the implication of (BO+) further below, as (BO+) also makes its appearance in the context of another principle that has been used to discredit bow-tie views.

[4] Assume $\mathrm{E!}m \,\&\, \mathrm{E!}n \,\&\, \mathrm{S}m \,\&\, \mathrm{S}n$. Assume for *reductio* $m$ PREC $n$. By ($\mathrm{P2_R}$), ($\mathrm{P3_R}$), ($\mathrm{A34_R}$) and ($\mathrm{A23_R}$), this yields $@n(\mathrm{E!}n \,\&\, \blacktriangle\neg\mathrm{E!}n \,\&\, \blacktriangledown\neg\mathrm{E!}n \,\&\, \neg\mathrm{E!}m)$. By ($\mathrm{T2_R}$), this yields $\bigcirc(\mathrm{E!}n \,\&\, \blacktriangle\neg\mathrm{E!}n \,\&\, \blacktriangledown\neg\mathrm{E!}n \,\&\, \neg\mathrm{E!}m)$. Given our initial assumption, we can derive $\neg\triangle(\mathrm{E!}n \,\&\, \blacktriangle\neg\mathrm{E!}n \,\&\, \blacktriangledown\neg\mathrm{E!}n \,\&\, \neg\mathrm{E!}m)$, $\neg\triangledown(\mathrm{E!}n \,\&\, \blacktriangle\neg\mathrm{E!}n \,\&\, \blacktriangledown\neg\mathrm{E!}n \,\&\, \neg\mathrm{E!}m)$ and $\neg(\mathrm{E!}n \,\&\, \blacktriangle\neg\mathrm{E!}n \,\&\, \blacktriangledown\neg\mathrm{E!}n \,\&\, \neg\mathrm{E!}m)$. This leaves as the only option $\triangleleft(\mathrm{E!}n \,\&\, \blacktriangle\neg\mathrm{E!}n \,\&\, \blacktriangledown\neg\mathrm{E!}n \,\&\, \neg\mathrm{E!}m)$ which, it would seem, can only be rejected if (BO+) is assumed.

[5] Suppose $\varphi \rightarrow \blacktriangleleft\varphi$ is a theorem. By (AR1) and (RR), then, $\triangleleft\varphi \rightarrow \triangleleft\blacktriangleleft\varphi$ is a theorem. Using (AR5) and again (AR1) and (RR), $\triangleleft\blacktriangleleft\varphi \rightarrow \varphi$ is a theorem. So, $\triangleleft\varphi \rightarrow \varphi$ is a theorem. Conversely, suppose the latter is a theorem. Then by (AR1) and (RR), so is $\blacktriangleleft\triangleleft\varphi \rightarrow \blacktriangleleft\varphi$. By (AR5), then, $\varphi \rightarrow \blacktriangleleft\varphi$ is a theorem.

Thus, in his critique of presentism in the context of SR, Putnam invokes the following principle:

> If it is the case that all and only the things that stand in a certain relation *R* to me-now are real, and you-now are also real, then it is also the case that all and only the things that stand in the relation *R* to you-now are real. (Putnam 1967: 241)

Transposed to the key of our current discussion, the principle may faithfully be recast as.

(4) $\mathrm{E!}x \,\&\, @x\mathrm{E!}y \rightarrow \mathrm{E!}y$

Given spatiopermanentism, (4) trivially holds. The proof that (4) also holds on pointy spatiopresentism, and on pointy relativistic GBT, is straightforward:

*Proof:* Pointy spatiopresentism: Assume both $\mathrm{E!}x$ and $@x\mathrm{E!}y$. By ($\mathrm{P2_R}$), ($\mathrm{P3_R}$), (PO) and ($\mathrm{A23_R}$), the latter yields: $@x(\mathrm{E!}x \,\&\, \blacktriangle\neg\mathrm{E!}x \,\&\, \blacktriangledown\neg\mathrm{E!}x \,\&\, \blacktriangleleft\neg\mathrm{E!}x \,\&\, \mathrm{E!}y)$. By ($\mathrm{T2_R}$), $\bigcirc(\mathrm{E!}x \,\&\, \blacktriangle\neg\mathrm{E!}x \,\&\, \blacktriangledown\neg\mathrm{E!}x \,\&\, \blacktriangleleft\neg\mathrm{E!}x \,\&\, \mathrm{E!}y)$. Given $\mathrm{E!}x$, $\mathrm{E!}y$. Pointy relativistic GBT: Assume both $\mathrm{E!}x$ and $@x\mathrm{E!}y$. By ($\mathrm{P2_R}$), (PO) and ($\mathrm{A23_R}$), the latter yields: $@x(\mathrm{E!}x \,\&\, \blacktriangle\neg\mathrm{E!}x \,\&\, \blacktriangleleft\neg\mathrm{E!}x \,\&\, \mathrm{E!}y)$. By ($\mathrm{T2_R}$), $\bigcirc(\mathrm{E!}x \,\&\, \blacktriangle\neg\mathrm{E!}x \,\&\, \blacktriangleleft\neg\mathrm{E!}x \,\&\, \mathrm{E!}y)$. Given $\mathrm{E!}x$ and ($\mathrm{P1_R}$), $\mathrm{E!}y$. □

On bow-tie views, by contrast, (4) can only be derived in analogous ways on condition that (BO+) holds. Thus, assume bow-tie spatiopresentism and assume both $\mathrm{E!}x$ and $@x\mathrm{E!}y$. By ($\mathrm{P2_R}$), ($\mathrm{P3_R}$), (BO) and ($\mathrm{A23_R}$), the latter yields: $@x(\mathrm{E!}x \,\&\, \blacktriangle\neg\mathrm{E!}x \,\&\, \&\, \blacktriangledown\neg\mathrm{E!}x \,\&\, \blacktriangleleft\mathrm{E!}x \,\&\, \mathrm{E!}y)$. By ($\mathrm{T2_R}$), $\bigcirc(\mathrm{E!}x \,\&\, \blacktriangle\neg\mathrm{E!}x \,\&\, \blacktriangledown\neg\mathrm{E!}x \,\&\, \blacktriangleleft\mathrm{E!}x \,\&\, \mathrm{E!}y)$. Given $\mathrm{E!}x$, either $(\mathrm{E!}x \,\&\, \blacktriangle\neg\mathrm{E!}x \,\&\, \blacktriangledown\neg\mathrm{E!}x \,\&\, \blacktriangleleft\mathrm{E!}x \,\&\, \mathrm{E!}y)$ or $\triangleleft(\mathrm{E!}x \,\&\, \blacktriangle\neg\mathrm{E!}x \,\&\, \blacktriangledown\neg\mathrm{E!}x \,\&\, \blacktriangleleft\mathrm{E!}x \,\&\, \mathrm{E!}y)$. In the first case, $\mathrm{E!}y$. But in order to be able to derive $\mathrm{E!}y$ in the second case, we would need '$\triangleleft\mathrm{E!}y \rightarrow \mathrm{E!}y$', which is equivalent to (BO+).

Alternatively, assume bow-tie relativistic GBT and assume both $\mathrm{E!}x$ and $@x\mathrm{E!}y$. By ($\mathrm{P2_R}$), (BO) and ($\mathrm{A23_R}$), the latter yields: $@x(\mathrm{E!}x \,\&\, \blacktriangle\neg\mathrm{E!}x \,\&\, \blacktriangleleft\mathrm{E!}x \,\&\, \mathrm{E!}y)$. By ($\mathrm{T2_R}$), $\bigcirc(\mathrm{E!}x \,\&\, \blacktriangle\neg\mathrm{E!}x \,\&\, \blacktriangleleft\mathrm{E!}x \,\&\, \mathrm{E!}y)$. Given $\mathrm{E!}x$, either $(\mathrm{E!}x \,\&\, \blacktriangle\neg\mathrm{E!}x \,\&\, \blacktriangleleft\mathrm{E!}x \,\&\, \mathrm{E!}y)$ or $\triangle(\mathrm{E!}x \,\&\, \blacktriangle\neg\mathrm{E!}x \,\&\, \blacktriangleleft\mathrm{E!}x \,\&\, \mathrm{E!}y)$ or $\triangleleft(\mathrm{E!}x \,\&\, \blacktriangle\neg\mathrm{E!}x \,\&\, \blacktriangleleft\mathrm{E!}x \,\&\, \mathrm{E!}y)$. In the first two cases, given ($\mathrm{P1_R}$), $\mathrm{E!}y$. But in order to be able to derive $\mathrm{E!}y$ in the third case, we would need '$\triangleleft\mathrm{E!}y \rightarrow \mathrm{E!}y$', which is equivalent to (BO+).

Accordingly, like the (PO)-independent proof of (3) – the principle we extracted from Savitt (2000) – the (PO)-independent proof of (4) relies on

(BO+) $\mathrm{E!}x \rightarrow \blacktriangleleft\mathrm{E!}x$

But now, as long as we limit the range of viable views to views whose ontological commitments do not trace any more fine-grained divisions than those the fourfold causal structure of spacetime permits, (BO+) can easily be seen to collapse into (SPER). It is clear that (BO+) fails on the pointy views above – and similarly on the pointy view, not discussed here, according to which only the here-now and the causal future of here-now are populated: (BO+) together with any pointy view yields $@x\blacktriangleleft(\mathrm{E!}x \,\&\, \neg\mathrm{E!}x)$, which corresponds to the fact that the relation of causal

precedence between spacetime-points is a total relation. Using ($A22_R$) it can indeed be shown that if we have $@x\blacktriangleleft(E!x \,\&\, \neg E!x)$ as a theorem, then $\neg\triangleleft\varphi$ is a theorem, and so, ($A5_R$) and ($A6_R$) boil down to the usual past- and future-linearity principles. The upshot is that, granted that we are in a genuinely relativistic setting, (BO+) fails on pointy views.

On bow-tie views, as Savitt notes, there are points $s$ and $s'$ in the elsewhere region of here-now such that $s$ is in the causal future of $s'$. Accordingly, if at $s$, $s$ did not exist anywhere in the causal past of $s$ – as per ($P2_R$) it did not – then $s$ cannot exist at $s'$ – contrary to what (BO+) would demand. (BO+) is similarly at odds with the bow-tie view, not discussed here, according to which only the causal past is unpopulated: if at $s'$, $s'$ will not exist anywhere in the causal future of $s'$, then $s'$ cannot exist at $s$ – contrary to what (BO+) demands. The only principled option left is spatiopermanentism.

Another way of arguing that, in a relativistic setting, (BO+) is in tension with spatiotemporaryism quite generally, is this. Consider the converse of (AR2), which is equivalent to $\blacktriangleleft\blacktriangleleft\varphi \rightarrow \bullet\varphi$. Given (BO+), and given that $\blacktriangleleft$ behaves normally, $\blacktriangleleft E!x \rightarrow \blacktriangleleft\blacktriangleleft E!x$ holds, and hence, again by (BO+), so does $E!x \rightarrow \blacktriangleleft\blacktriangleleft E!x$. By the converse of (AR2), we infer $E!x \rightarrow \bullet E!x$, which, by ($A16_R$) and the fact that $\bullet$ is an S5 modality, yields $\bullet E!x$, and hence spatiopermanentism. In a relativistic setting, the converse of (AR2) looks very weak; it anyway holds in standard Minkowskian spacetime, and presumably in many 'well-behaved' spacetimes conforming to GR. Insofar as in combination with (BO+), the converse of (AR2) yields spatiopermanentism, spatiotemporaryists should refuse taking (BO+) on board.

Accordingly, pointy spatiopresentists can accept both (3) and (4), but must reject (BO+). Proponents of pointy relativistic GBT can accept (4), but must reject (3) and (BO+). Bow-tie spatiopresentists and proponents of bow-tie relativistic GBT must reject all of (3), (4) and (BO+). It is true that rejection of (4) implies a more thoroughgoing departure from the prerelativistic counterparts of either of the latter types of view. For, in a prerelativistic setting,

$$E!x \,\&\, (\text{At } x, E!y) \rightarrow E!y$$

which is the principle corresponding to (4), is easily proved to hold on any of the classical views.

*Proof:* (i) Permanentism: On permanentism, $E!y$ anyway holds. (ii) GBT: Assume both $E!x$ and At $x$, $E!y$. Given (P2) and (A23), the latter yields: At $x$, $(E!x \,\&\, H\neg E!x \,\&\, E!y)$. By (T2), from this we derive: Sometimes, $(E!x \,\&\, H\neg E!x \,\&\, E!y)$. Since $E!x$, either $(E!x \,\&\, H\neg E!x \,\&\, E!y)$ or $P(E!x \,\&\, H\neg E!x \,\&\, E!y)$, and hence either $E!y$ or $PE!y$. Either way, given (P1), $E!y$ follows. (iii) Presentism: Assume both $E!x$ and At $x$, $E!y$. Given (P3), (P2) and (A23), the latter yields: At $x$, $(E!x \,\&\, H\neg E!x \,\&\, G\neg E!x \,\&\, E!y)$. By (T2), from this derive: Sometimes, $(E!x \,\&\, H\neg E!x \,\&\, G\neg E!x \,\&\, E!y)$. Since $E!x$, $E!y$ follows. □

Similarly, classical presentism vindicates

$$\mathrm{E}!x \,\&\, \mathrm{E}!y \,\&\, \mathrm{T}x \,\&\, \mathrm{T}y \rightarrow \neg(x \prec y)$$

which corresponds to (3). The only candidate prerelativistic principle corresponding to (BO+) would be the present-tensed tautology '$\mathrm{E}!x \rightarrow \mathrm{E}!x$', if only because, in the prerelativistic setting, all non-causally separated points, or events, are contemporaneous. To the extent that it is uncontroversial that in a relativistic setting, all spatiotemporaryist views must reject (BO+) and there is no objective sense in which non-causally separated points, or events, are contemporaneous, it should not be surprising that some of the counterparts of other prerelativistic principles must, in such a setting, likewise be given up – especially if, as argued here, vindication of those counterpart principles would, within the context of certain theoretical choices, require acceptance of (BO+).

Accordingly, it is far from clear why bow-tie spatiotemporaryism's rejection of (4) should be held against it, solely on the grounds that its prerelativistic precursors adopted its prerelativistic counterpart. Again, the appeal to everyday intuition has little to recommend itself, if only because the idea that there is a region of points non-causally separated from here-now some of which stand in relations of causal precedence to one another is itself not very intuitive by everyday standards.

We conclude that the bow-tie versions of spatiotemporaryism – not only are superior to their pointy rivals, in that unlike the latter, they can account for the persistence of non-causally separated objects of which we have perceptual knowledge – but also prove quite resilient in the face of prominent, principle-driven attempts to discredit them, as no convincing case has been made for thinking that spatiotemporaryists must show allegiance to the principles invoked. That bow-tie versions of spatiotemporaryism are the most natural candidates for serving as the relativistic counterparts to classical presentism and classical GBT, respectively, will independently be confirmed in the next and last section of this chapter.

## 9.5 Recapture of the Classical Views

We have hitherto been concerned with the articulation of relativistic theories of spacetime that promise to be natural counterparts of prerelativistic theories of time. However, so far, we have left the relation between relativistic theories and their prerelativistic counterparts at a somewhat intuitive level. As advertised towards the end of Chap. 8, to complete implementation of the revisionary strategy, we must find a principle, or conjunction of principles, $q$, couched in relativistic terms, such that for any prerelativistic theory $T$, its relativistic counterpart $T_R$ implies $T$ on condition that $q$ holds, where $q$, though consistent with $T_R$, is inconsistent with rejection of an absolute and total temporal order. We may then conceive of $T$ as being decomposable into two independent parts, $q$ and $T_R$, such that even if $q$ fails, $T_R$ may nonetheless survive relativity.

The distinction between pointy and bow-tie versions of spatiotemporaryism might suggest that one such prerelativistic theory $T$ can have more than one relativistic counterpart, so that the decomposition might not be unique, which is unproblematical as long as, given $q$, these relativistic counterparts collapse into one another. However, as we shall see, such a collapse is not to be had and bow-tie views prove the by far more natural candidates for being relativistic counterparts of classical versions of temporaryism.

The search for such a $q$ must be guided by the following thought. Prerelativistic physics presumes that there are spatially separated events that are absolutely simultaneous with one another, and so in effect, that there are distinct spacetime-points which share the same absolute time coordinate. The principle $q$ should accordingly allow us to conceive of absolute times as corresponding to equivalence classes of spacetime-points such that anywhere any spacetime-point non-causally separated from, or identical with, some member of such a class *eo ipso* belongs to that same class. This *inter alia* requires that, given $q$, the relation SIM should prove to be an equivalence relation:

$$x \text{ SIM } y \equiv_{df} \text{S}x \,\&\, \text{S}y \,\&\, \neg(x \text{ PREC } y) \,\&\, \neg(y \text{ PREC } x)$$

Similarly, $q$ should allow us to conceive of relations of precedence among absolute times as corresponding to suitable relations among such equivalence classes, and of relations of absolute temporal location in terms of suitable relations of being somewhere located at some member of such a class.

These are not the only constraints on $q$ that should guide us, however. In prerelativistic theories of time, with their absolute conception of tense, what is the case at most varies across time, but not across space. Accordingly, a further thought must be that, given $q$, whatever is the case at any one member of such an equivalence class, likewise holds at any other member of that class.

In this spirit, we now suggest that the sought-after $q$ should be conceived of as the conjunction of the following two bridge principles:

(PR1) $x \text{ PREC } y \,\&\, x \text{ SIM } z \rightarrow z \text{ PREC } y$
(PR2) $@x\varphi \,\&\, x \text{ SIM } y \rightarrow @y\varphi$

In Appendix 3, we shall provide all the nitty-gritty details of how the combination of these bridge principles successfully serves its purpose. Here, we merely sketch some central features. Thus, note that (PR1) straightforwardly implies that SIM is transitive. Consequently, the bow-tie region of each spacetime-point, comprising that point and its elsewhere region, shrinks, as it were, to a single hypersurface. Given that we anyway have both '$\text{S}x \rightarrow x \text{ SIM } x$' and '$x \text{ SIM } y \rightarrow y \text{ SIM } x$', (PR1) accordingly ensures that SIM is an equivalence relation, which is just as needed. It should also be clear that in a relativistic setting, (PR1) fails, precisely because the bow-tie region then has depth.

(PR2) expresses that there is no factual variation across members of the same equivalence class of spacetime-points, which is also just as needed. (PR2) implies both of the following:

(TR1) $\triangleleft\varphi \rightarrow \varphi$
(TR2) $\varphi \,\&\, \triangleleft\top \rightarrow \triangleleft\varphi$ (with $\top$ an arbitrary tautology)

*Proof of (TR1):* Let $x$ be a variable that is not free in $\varphi$. From (AR6), i.e.

$$@x\blacktriangleleft\varphi \leftrightarrow \mathrm{S}x \,\&\, \bullet\forall y(y \text{ SEP } x \rightarrow @y\varphi)$$

we get $@x\triangleleft\varphi \rightarrow \bigcirc\exists y(y \text{ SEP } x \,\&\, @y\varphi)$. Given (PR2) we then get $@x\triangleleft\varphi \rightarrow @x\varphi$, and then $\mathrm{S}x \rightarrow @x(\triangleleft\varphi \rightarrow \varphi)$, and then $\forall x(\mathrm{S}x \rightarrow @x(\triangleleft\varphi \rightarrow \varphi))$. Using $(\mathrm{A22_R})$, i.e.

$$\forall x(\mathrm{S}x \rightarrow @x\varphi) \rightarrow \varphi \quad \text{(with } x \text{ not free in } \varphi\text{)}$$

one can then derive $\triangleleft\varphi \rightarrow \varphi$. □

*Proof of (TR2):* From (TR1), we get $\varphi \rightarrow \blacktriangleleft\varphi$, and hence $\varphi \rightarrow \blacktriangleleft(\top \rightarrow \varphi)$. (TR2) follows, since we have in general $\blacktriangleleft(\psi \rightarrow \xi) \rightarrow (\triangleleft\psi \rightarrow \triangleleft\xi)$. □

It should be clear that, at least from a spatiotemporaryist perspective, neither (TR1) nor (TR2) holds in a relativistic setting. By contrast, in a prerelativistic setting – where the bow-tie region of each spacetime-point shrinks to a single hypersurface, and where there is assumed to be no factual variation across points on such a hypersurface – both (TR1) and (TR2) should hold.

In order to derive prerelativistic views from their relativistic counterparts *modulo* our bridge principles (PR1) and (PR2), we first systematically enrich the relativistic spacetime-point-sensitive language by a dyadic predicate $\varepsilon$ for class membership, as well as a denumerably infinite stock of variables distinct from the variables of the original language. Since we are only interested in classes of spacetime-points, $\varepsilon$ is understood to be restricted to membership in such classes. We adopt the following definition of the predicate C for classes:

$$\mathrm{C}m \equiv_{\mathrm{df}} \bigcirc\exists x(x\varepsilon m) \quad \text{(with } x \text{ the first new variable distinct from } m\text{)}$$

Of all the axioms governing $\varepsilon$ which we mention in Appendix 3, we here merely highlight the following comprehension axiom:

(AX6) $\mathrm{S}x \rightarrow @x\exists y\bullet\forall z(z\,\varepsilon\, y \leftrightarrow z \text{ SIM } x \,\&\, @x\mathrm{E}!z))$

We next define:

- $\mathrm{SC}m \equiv_{\mathrm{df}} \mathrm{C}m \,\&\, \bullet\forall x\bullet\forall y(x\,\varepsilon\, m \,\&\, y\,\varepsilon\, m \rightarrow x \text{ SIM } y)$
  (with $x$ and $y$ respectively the first and second new variable distinct from $m$)
- $m \text{ SIMEM } n \equiv_{\mathrm{df}} \bigcirc\exists x(m \text{ SIM } x \,\&\, x\,\varepsilon\, n)$
  (with $x$ the first new variable distinct from both $m$ and $n$)

Here 'SC' can be read as being short for 'is a simultaneity class' and 'SIMEM' as being short for 'is simultaneous with a member of'.

Armed with these definitions, and on assumption of (PR1) and (PR2), we can now define notions that respectively correspond to the prerelativistic notions of a time-instant, temporal precedence among such time-instants, location at a time-instant, and holding at a time-instant:

- $T^*m \equiv_{df} SCm \,\&\, \bullet\forall x(x \text{ SIMEM } m \rightarrow x \,\varepsilon\, m)$
- $m \prec^* n \equiv_{df} T^*m \,\&\, T^*n \,\&\, \bullet\forall x\bullet\forall y(x \,\varepsilon\, m \,\&\, y \,\varepsilon\, n \rightarrow x \text{ PREC } y)$
- $m \,L^*\, n \equiv_{df} T^*n \,\&\, \bigcirc\exists x(x \,\varepsilon\, n \,\&\, m \text{ LOC } x)$
- $At^*\, m, \varphi \equiv_{df} T^*m \,\&\, \bullet\forall x(x \,\varepsilon\, m \rightarrow @x\varphi)$

In the presence of (PR1) and (PR2), T*, $\prec^*$, L* and At* can indeed serve as relativistic translations of the prerelativistic T, $\prec$, L and At. A detailed translation manual taking the prerelativistic language into the relativistic language can be found in Appendix 3. However, we note already here that ▲ and ▼ serve as natural translations of H and G, respectively.

We can now establish the following theorems:

(A30#) $\neg(x \prec^* x)$
(A31#) $(x \prec^* y \,\&\, y \prec^* z) \rightarrow x \prec^* z$
(A32#) $(T^*x \,\&\, T^*y) \rightarrow (x \prec^* y \vee x = y \vee y \prec^* x)$

(A30#) already follows from the aforementioned definitions, while (A31#) can be established using ($A31_R$), i.e. '$(x \text{ PREC } y \,\&\, y \text{ PREC } z) \rightarrow x \text{ PREC } z$'. The totality principle (A32#), by contrast, depends on (PR1). (A30#) to (A32#) mimic the classical axioms (A30) to (A32) which latter encode central features of the absolute and total temporal order that is characteristic of prerelativistic physics.

The systematic recapture of the other prerelativistic axioms, including those for temporal location, can be found in Appendix 3. There, we will give a translation of the prerelativistic language into the enriched relativistic language and show in detail that in the presence of our bridge principles, for any theorem of the classical neutral system described in Appendix 1, its translation is a theorem of the relativistic neutral system described in Appendix 2, and for any theorem of classical GBT/presentism/permanentism, its translation is a theorem of relativistic GBT/presentism/permanentism.

Before closing, let us here briefly review how the more specific principles introduced in the previous section fare, once (PR1) and (PR2) are added to the stock. We begin by considering the characteristic principles for pointy and for bow-tie views, i.e.

(PO) $Sx \rightarrow @x\blacktriangleleft\neg E!x$
(BO) $Sx \rightarrow @x\blacktriangleleft E!x$

(BO) already follows from (TR1) and ($A23_R$), i.e. '$Sx \rightarrow @xE!x$'. If (PO) is accepted, then given (BO) and (AR6), one can derive:

(2) $Sx \rightarrow \bullet\neg\exists y(x \text{ SEP } y)$

which latter is equivalent to:

– $Sx \rightarrow \bullet\forall y(x \text{ SIM } y \rightarrow x = y)$

This shows very clearly that pointy views are implausible candidates for being the relativistic counterparts of classical temporaryist views: there is no suggestion in classical presentism that space shrinks to a single point, and no suggestion in classical GBT either that the edge of becoming is not really an edge but a point. This leaves bow-tie spatiopresentism and bow-tie relativistic GBT as the only serious contenders. As argued in the previous section, in a relativistic setting, both these views must reject.

(BO+) $E!x \rightarrow \blacktriangleleft E!x$

By contrast, we argued that in such a relativistic setting, the converse of (AR2), i.e.

(C-AR2) $\bigcirc\varphi \rightarrow \triangleleft\triangleleft\varphi$

is eminently plausible. This assessment is reversed in a prerelativistic setting. Thus, once (PR2) is added to the mix, (TR1) becomes derivable, and (TR1) straightforwardly implies (BO+). However, given that same theorem (TR1), (C-AR2) straightforwardly implies $\bigcirc\varphi \rightarrow \varphi$, and so in particular, $\triangle\varphi \rightarrow \varphi$ and $\triangledown\varphi \rightarrow \varphi$. None of these latter principles is acceptable to spatiotemporaryists, irrespective of whether (PR1) and (PR2) are assumed to hold.

Rejecting (C-AR2) while accepting (BO+) thus becomes mandatory for spatiotemporaryists, whether they defend pointy or bow-tie views, once we change from a relativistic to a prerelativistic setting, in which latter setting both our bridge principles hold. Note that this reversal in no way undermines the contention that, in the presence of (PR1) and (PR2), for any theorem of prerelativistic views, its relativistic translation is a theorem of their respective relativistic counterparts. For, neither (C-AR2) nor the negation of (BO+) is a theorem of the relevant relativistic views taken on their own, i.e. without making further assumptions about whether or not we find ourselves in a relativistic setting in which the elsewhere region has depth.

We similarly noted in the previous section that in a relativistic setting, bow-tie spatiopresentists and proponents of bow-tie relativistic GBT alike must reject both of the following:

(3) $E!x \,\&\, E!y \,\&\, Sx \,\&\, Sy \rightarrow \neg(x \text{ PREC } y)$
(4) $E!x \,\&\, @xE!y \rightarrow E!y$

In a prerelativistic setting, however, in which both (PR1) and (PR2) hold, bow-tie spatiopresentists will accept both (3) and (4), alongside their pointy friends, while proponents of bow-tie relativistic GBT, just like *their* pointy friends, will continue rejecting (3), while now accepting (4). Again, these findings in no way challenge our claim that, in the presence of (PR1) and (PR2), for any theorem of prerelativistic

views, its relativistic translation is a theorem of their respective relativistic counterparts.

This concludes our discussion of the revisionary strategy and the metaphysical views its implementation allows us to articulate (however, see Appendix 2 and Appendix 3 for a more thoroughgoing formal treatment). In the previous chapter we conceded to the friends of the conservative strategy that it is an open empirical question whether physics may ultimately reinstate assumption of an absolute and total temporal order. So, in the end there may prove to be no need to contemplate the different spatiotemporaryist views expounded here. However, it was part of the rationale of the revisionary strategy that we should not make our metaphysical options hostage to such empirical fortune and prepare for the case in which the relativistic conception of spacetime persists. Although we do not pretend to have given knock-down arguments against pointy versions of spatiotemporaryism, and have not even begun to negotiate between the two bow-tie versions introduced above, the mere provision of a theoretical and logical framework in which to articulate and discuss such issues seems to us to be a major advance in the metaphysical study of (space)time.